AcademyCloudFoundation...
Presentation

Make notes Very detailed (full sentences, no short phrases)
✅ Including all important theoretical points
✅ Including procedural steps wherever steps are given
✅ Formatted cleanly for exam
make sure to include each and every imp point.

Below are **very detailed, full-sentence, cleanly formatted exam-ready notes** from **Module 10: Automatic Scaling and Monitoring – AWS Academy Cloud Foundations**, covering **every important theoretical and procedural point** from the provided material

AcademyCloudFoundations_Module_...

📘 MODULE 10: AUTOMATIC SCALING AND MONITORING (AWS)

🔹 MODULE OVERVIEW

This module focuses on how AWS automatically adjusts computing resources to handle changing workloads and how performance is continuously monitored.

The main topics covered are:

• Elastic Load Balancing
• Amazon CloudWatch
• Amazon EC2 Auto Scaling

The module also includes hands-on activities and a lab to practically implement scaling and monitoring.

🎯 MODULE OBJECTIVES

After completing this module, a learner should be able to:

1. Explain how traffic is distributed across multiple Amazon EC2 instances using Elastic Load Balancing.
2. Describe how Amazon CloudWatch monitors AWS resources and applications in real time.
3. Explain how Amazon EC2 Auto Scaling automatically launches and terminates instances based on workload demand.
4. Perform architectural improvements using scaling and load balancing techniques.

📈 WHY SCALING IS IMPORTANT

Scaling is important because:

• Application traffic is not constant and changes throughout the day, week, and year.
• Without scaling, systems may crash during high demand or waste money during low usage.
• Scaling ensures high availability, better performance, and cost efficiency.

Real-world example:
Traffic on Amazon.com increases significantly during special events like November sales and holidays, requiring automatic capacity increases.

🔁 AMAZON EC2 AUTO SCALING

Amazon EC2 Auto Scaling is a service that:

• Maintains application availability automatically
• Adds EC2 instances when demand increases
• Removes EC2 instances when demand decreases
• Replaces unhealthy or impaired instances automatically

✅ KEY BENEFITS

Amazon EC2 Auto Scaling:

✔ Ensures applications remain available
✔ Adjusts capacity automatically
✔ Reduces manual management
✔ Saves cost by running only required resources

📊 SCALING OPTIONS IN AUTO SCALING

There are several scaling methods:

1. Manual Scaling

You manually increase or decrease the number of EC2 instances.

2. Scheduled Scaling

Scaling occurs at specific times (for example, more servers during business hours).

3. Dynamic (On-Demand) Scaling

Scaling automatically occurs based on CloudWatch metrics such as CPU usage.

4. Predictive Scaling

AWS forecasts traffic patterns and scales resources in advance.

📦 AUTO SCALING GROUPS

An **Auto Scaling Group (ASG)** is:

➡ A logical collection of EC2 instances
➡ Treated as a single unit for scaling and management

Purpose of ASG:

• Defines minimum number of instances
• Defines maximum number of instances
• Maintains desired number of running instances
• Automatically replaces failed instances

➕ SCALING OUT vs ➖ SCALING IN

🔼 Scaling Out:

Adding more EC2 instances to handle increased load.

🔽 Scaling In:

Removing EC2 instances when demand decreases.

⚙ HOW AMAZON EC2 AUTO SCALING WORKS (STEP BY STEP)

1. You create a **launch configuration** (or launch template) that defines:
    - Instance type
    - AMI
    - Security groups
    - Key pairs
2. You create an **Auto Scaling group** using that configuration.
3. You set:
    - Minimum capacity
    - Maximum capacity
    - Desired capacity
4. You configure **scaling policies** based on CloudWatch metrics.
5. Amazon CloudWatch monitors performance.
6. When thresholds are crossed:
    - New instances are launched (scale out)
    - Instances are terminated (scale in)

📉 IMPLEMENTING DYNAMIC SCALING

Dynamic scaling works using three main services:

✔ Amazon EC2 Auto Scaling

Handles adding/removing instances.

✔ Amazon CloudWatch

Monitors metrics such as CPU utilization, network traffic, and memory usage.

✔ Elastic Load Balancing

Distributes traffic evenly across running instances.

🌐 AWS AUTO SCALING (SERVICE)

AWS Auto Scaling is a **separate service** that manages scaling across multiple AWS resources.

It:

• Monitors applications automatically
• Maintains performance at the lowest possible cost
• Uses a unified interface for scaling plans

🔧 RESOURCES SUPPORTED BY AWS AUTO SCALING

It can scale:

• Amazon EC2 instances and Spot Fleets
• Amazon ECS tasks
• Amazon DynamoDB tables and indexes
• Amazon Aurora replicas

📌 SECTION 3 – KEY TAKEAWAYS

• Scaling helps quickly respond to changing resource demands.
• Amazon EC2 Auto Scaling automatically adjusts EC2 capacity.
• An Auto Scaling group is a logical set of EC2 instances.
• A launch configuration defines how new instances are created.
• Dynamic scaling uses Auto Scaling, CloudWatch, and Load Balancing together.
• AWS Auto Scaling is different from EC2 Auto Scaling.

🧪 LAB 6: SCALE AND LOAD BALANCE YOUR ARCHITECTURE

📖 LAB SCENARIO

The lab demonstrates how to:

- Create scalable infrastructure
- Balance traffic
- Monitor system performance

📝 LAB TASKS (PROCEDURAL STEPS)

Step 1: Create an AMI

You create an Amazon Machine Image from a running EC2 instance to use as a template.

Step 2: Create an Application Load Balancer

This distributes incoming traffic across multiple instances.

Step 3: Create a Launch Configuration

This defines instance settings like AMI, instance type, and security groups.

Step 4: Create an Auto Scaling Group

This launches instances automatically inside a private subnet.

Step 5: Configure CloudWatch Alarms

You set alarms to monitor performance metrics.

Step 6: Monitor Infrastructure

You observe scaling actions in real time.

🎯 LAB OUTCOME

A fully automated architecture that:

✔ Automatically scales
✔ Balances traffic
✔ Monitors performance

📚 MODULE SUMMARY

By the end of this module, you learned:

- How Elastic Load Balancing distributes traffic across EC2 instances
- How Amazon CloudWatch provides real-time monitoring
- How Amazon EC2 Auto Scaling adjusts resources automatically
- How scaling improves reliability and performance

❓ SAMPLE EXAM QUESTION

Which service would you use to send alerts based on Amazon CloudWatch alarms?

Correct Answer:

✅ Amazon Simple Notification Service (SNS)

Explanation:

CloudWatch detects conditions, and SNS sends notifications such as emails or SMS alerts.

✅ QUICK REVISION POINTS (EXAM GOLD)

✔ Auto Scaling maintains availability automatically
✔ Scaling out = add instances
✔ Scaling in = remove instances
✔ ASG = group of EC2 instances
✔ CloudWatch monitors metrics
✔ Load Balancer distributes traffic
✔ AWS Auto Scaling handles multiple services

If you'd like, I can:

📌 Turn these into **short revision notes**
📌 Create **memory tricks**
📌 Make **important exam Q&A**
📌 Provide **diagram explanations**

Just tell me 👍

Sources

just write a good note on auto scaling

Here is a **well-structured, full-sentence, exam-ready detailed note on Auto Scaling** — written cleanly, with all important theoretical and procedural points included.

📈 AMAZON EC2 AUTO SCALING

Amazon EC2 Auto Scaling is an AWS service that automatically adjusts the number of Amazon EC2 instances running in an application based on workload demand. Its main purpose is to maintain application availability, improve performance, and reduce cost by ensuring that the correct amount of computing resources is always available.

Auto Scaling continuously monitors applications and automatically adds new EC2 instances when demand increases and removes instances when demand decreases. It also detects unhealthy or impaired instances and replaces them automatically without requiring manual intervention.

🔧 OBJECTIVES OF AUTO SCALING

The primary objectives of Amazon EC2 Auto Scaling are:

• To ensure high availability of applications
• To automatically respond to traffic fluctuations
• To reduce downtime by replacing failed instances
• To optimize costs by avoiding over-provisioning

📦 AUTO SCALING GROUP (ASG)

An Auto Scaling Group is a logical collection of EC2 instances that are managed together for scaling and availability purposes.

Each Auto Scaling Group defines:

• Minimum number of instances that must always be running
• Maximum number of instances allowed
• Desired number of instances to maintain under normal conditions

If an instance fails or becomes unhealthy, the Auto Scaling Group automatically launches a new instance to replace it.

⚙ LAUNCH CONFIGURATION (OR LAUNCH TEMPLATE)

A launch configuration is a template that defines how new EC2 instances are created. It includes:

• Amazon Machine Image (AMI)
• Instance type
• Security groups
• Key pair
• Storage configuration

Whenever Auto Scaling needs to add new instances, it uses this configuration.

➕ SCALING OUT AND ➖ SCALING IN

Scaling Out refers to increasing the number of EC2 instances to handle increased workload or traffic.

Scaling In refers to decreasing the number of EC2 instances when workload demand reduces.

📊 SCALING POLICIES

Auto Scaling uses scaling policies to decide when to add or remove instances. These policies are generally based on performance metrics monitored by Amazon CloudWatch.

Common metrics include:

• CPU utilization
• Network traffic
• Memory usage
• Request count

For example, if CPU usage exceeds a specified threshold, Auto Scaling automatically launches new instances.

🔁 TYPES OF SCALING

Manual Scaling

The user manually changes the number of running instances.

Scheduled Scaling

Scaling occurs at predefined times, such as increasing capacity during business hours.

Dynamic Scaling

Scaling automatically happens based on real-time CloudWatch metrics.

Predictive Scaling

AWS forecasts future demand and scales resources in advance.

🔄 WORKING OF AUTO SCALING (STEP-BY-STEP)

1. A launch configuration is created defining instance settings.
2. An Auto Scaling Group is created with minimum, maximum, and desired capacity.
3. CloudWatch monitors performance metrics continuously.
4. Scaling policies are applied.
5. When thresholds are crossed:
    - Instances are added (scale out)
    - Instances are removed (scale in)
6. Unhealthy instances are automatically replaced.

🌟 ADVANTAGES OF AUTO SCALING

• Maintains consistent application performance

• Automatically handles traffic spikes

• Reduces operational overhead

• Improves fault tolerance

• Minimizes cost

📌 CONCLUSION (Optional for exams)

Amazon EC2 Auto Scaling is a powerful AWS service that ensures applications run efficiently by automatically adjusting resources based on real-time demand. It plays a critical role in building highly available, scalable, and cost-effective cloud architectures.

If you want, I can:

📘 Convert this into **5-mark or 10-mark exam format**

🧠 Add **memory tricks**

📊 Provide **diagram explanation of Auto Scaling flow**

Just tell me!

5 mark ans on auto scaling, in points with full sentences

Here is a **perfect 5-mark exam-oriented answer on Auto Scaling**, written **in clear points with full sentences** (no short phrases), exactly the way evaluators prefer:

📈 AMAZON EC2 AUTO SCALING (5 MARKS)

1. Amazon EC2 Auto Scaling is an AWS service that automatically increases or decreases the number of EC2 instances based on workload demand to maintain application availability and performance.
2. It continuously monitors system performance using Amazon CloudWatch metrics such as CPU utilization and network traffic to decide when scaling actions are required.
3. Auto Scaling uses an Auto Scaling Group, which is a logical collection of EC2 instances that defines the minimum, maximum, and desired number of instances.
4. When traffic increases, Auto Scaling launches new instances automatically (scaling out), and when traffic decreases, it terminates unnecessary instances (scaling in).
5. It also detects unhealthy or failed EC2 instances and replaces them automatically, ensuring fault tolerance and reliability.

If you'd like, I can:

✔ Turn this into a **3-mark short answer**
✔ Make a **10-mark detailed answer**
✔ Add **diagram explanation in words**

Just say 👍